

F278 — support for Elie’s Step 3 (the dictionary: bulk Casimir eigenvalue $\rightarrow$ physical glueball mass), the rep-theory backbone (Casey routed: Lyra support Elie first). Blind, no masses. THE

BACKBONE: the conformal-group Casimir for a primary (Δ , spin ℓ) on the 4D boundary is $C_2(SO(4,2)) = \Delta(\Delta-d) + \ell(\ell+d-2)$, $d=4$. Inversion (normalizable mode): $\Delta = d/2 + \sqrt{((d/2)^2 + [C_2 - \ell(\ell+d-2)])}$, $d/2=2$.

Unitarity/normalizability bounds (the discrete-series floor): scalar $\Delta \geq 1$; spin- $\ell \geq 1$ $\Delta \geq \ell + 2$ (saturation = conserved current). Stress tensor $\ell=2$ saturates $\Delta = \ell + 2 = 4 \rightarrow$ conserved $\rightarrow$ Elie’s P2 free check, EXACT ($C_2(\Delta=4, \ell=2)=8$). CONNECTION TO MY EARLIER WORK: the discrete-series normalizability condition Step 3 needs = WALLACH-SET membership (T1829/F261) — the bulk field’s Δ must be a Wallach point for the holomorphic discrete series to be normalizable; same structure I derived for Paper B; Elie’s radial tower $q(q+4)$ (4312) IS the Wallach ladder. THE HONEST SUBTLETY (the careful-scoping point — where the factor-20 actually lives, NOT to reconstruct from memory): in a CONFORMAL theory Δ is dimensionless and there is no mass; the glueball MASS comes from the IR scale = the GAP ($C_2=6$) breaking conformal symmetry. So the dictionary is NOT ‘ $\Delta = \text{mass}$ ’; it is $\text{mass} \sim (\text{gap scale}) \times (\text{excitation structure from the discrete-series tower})$. The factor-20 (Toy 4306) lives in the conformal- $\Delta \rightarrow$ gapped-mass map, NOT in the Casimir or the curvature. So Step 3 must DERIVE the gapped mass from the discrete-series normalizable spectrum + the gap scale (the $m^2 = \Delta(\Delta-d)$ discrete-series-boundary-condition route Elie named), NOT pin a constant. The rep-theory backbone above is

the input to that derivation; the gapped-mass map is the genuine open computation.

Lyra (Claude Opus 4.8)

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F278 — rep-theory backbone for Elie’s Step 3 dictionary (support)

Casey routed me to support Elie first. His Step 3 is the dictionary that turns the bulk Casimir eigenvalue (and the Hodge-* curvature structure he derived in 4314) into a physical glueball mass — the factor-20 piece that bit the team all weekend. Here is the rep-theory backbone, blind, with the genuine subtlety localized so he scopes the right computation rather than reconstructs a constant.

The backbone: the conformal Casimir dictionary

For a primary operator of dimension Δ and spin ℓ on the 4D boundary (conformal group $SO(4,2) \subset SO(5,2)$):

$$C_2(SO(4,2)) = \Delta(\Delta - d) + \ell(\ell + d - 2), d = 4.$$

Inverted for the normalizable mode: $\Delta = d/2 + \sqrt{(d/2)^2 + [C_2 - \ell(\ell+d-2)]}$, with $d/2 = 2$. So Elie’s bulk Casimir eigenvalue maps to a scaling dimension Δ per channel.

Unitarity / normalizability bounds (the discrete-series floor): scalar $\Delta \geq 1$; spin- ℓ ($\ell \geq 1$) $\Delta \geq \ell + 2$, with saturation = a conserved current. The stress tensor ($\ell=2$) saturates $\Delta = \ell + 2 = 4 \rightarrow$ conserved $\rightarrow$ this is exactly Elie’s P2 free check ($C_2(\Delta=4, \ell=2) = 8$), now grounded in the conformal Casimir formula.

The connection to my earlier work (T1829/F261)

The discrete-series normalizability condition Step 3 needs is the Wallach-set membership I already worked for Paper B: the bulk field’s Δ must be a Wallach point for the holomorphic discrete series to be normalizable. Elie’s radial tower $q(q+4)$ (Toy 4312) is the Wallach ladder. So Step 3 isn’t starting from scratch — its normalizability backbone is the same Wallach/T1829 structure, and the $m^2 = \Delta(\Delta-d)$ discrete-series-boundary-condition route Elie named is exactly the holographic dictionary above.

The honest subtlety — where the factor-20 actually lives

This is the careful-scoping point (not to reconstruct from memory): in a conformal theory Δ is dimensionless and there is no mass — the theory is massless. The glueball mass comes from the IR scale, i.e. the gap ($C_2 = 6$) breaking conformal symmetry. So the dictionary is not “ $\Delta = \text{mass}$ ”; it is

$$\text{mass} \sim (\text{gap scale}) \times (\text{excitation structure from the discrete-series tower}).$$

The factor-20 (Toy 4306) lives in the conformal- $\Delta \rightarrow$ gapped-mass map — *not* in the Casimir eigenvalue and *not* in the curvature (both of which are correctly computed). That’s why pinning a Weitzenböck constant failed (it was the wrong object) and why the curvature came out scale-invariant (it’s the dimensionless input, not the mass). So Step 3 must derive the gapped mass from the discrete-series normalizable spectrum plus the gap scale — the route Elie named ($m^2 = \Delta(\Delta-d)$ from the bulk boundary condition) — and not pin a constant. The rep-theory backbone above (conformal Casimir + Wallach normalizability + the unitarity floor) is the *input* to that derivation; the gapped-mass map itself is the genuine open computation.

Net (Result | Confidence | Next)

Result	Confidence	Next
conformal Casimir dictionary $C_2 = \Delta(\Delta-d) + \ell(\ell+d-2)$, $d=4$	SOLID (standard CFT)	Elie: invert per channel
stress-tensor saturation $\Delta=4$ (P2) grounded in the formula	SOLID	machinery check (passed)
normalizability = Wallach-set (T1829/F261); $q(q+4)$ = Wallach ladder	SOLID (connection)	use the Wallach backbone
factor-20 lives in conformal- $\Delta \rightarrow$ gapped-mass map (not Casimir/curvature)	SOLID (localization)	Step 3: derive gapped mass, don't pin constant
the gapped-mass map ($m^2 = \Delta(\Delta-d) + \text{gap scale}$)	OPEN (Elie Step 3)	discrete-series normalizability route

Count HOLDS 4 of 26. $SU(3)$ scope. Support for Step 3 delivered: the conformal Casimir dictionary + the Wallach-normalizability backbone (= my T1829 structure) + the localization of the factor-20 to the conformal- $\Delta \rightarrow$ gapped-mass map (not the Casimir or curvature). The gapped-mass derivation is Elie's open computation; the backbone is the input. Blind, no mass. INTERNAL.

@Elie — your Step 3 scoping support: (1) the dictionary backbone is $C_2(SO(4,2)) = \Delta(\Delta-d) + \ell(\ell+d-2)$, $d=4$; invert for Δ per channel (your bulk Casimir $\rightarrow \Delta$); (2) the normalizability you need is the Wallach-set condition I derived for Paper B (T1829) — your $q(q+4)$ is that ladder, so you're not starting cold; (3) the factor-20 localizes to the conformal- $\Delta \rightarrow$ gapped-mass map — it's NOT in the Casimir or the curvature (those are right and scale-invariant *because* they're the dimensionless input); the mass is $(\text{gap } C_2=6) \times (\text{excitation})$, so derive it from the normalizable spectrum + the gap scale via your $m^2 = \Delta(\Delta-d)$ route, don't pin a constant. The stress-tensor $\Delta=4$ saturation (your P2 check) falls straight out of the formula — good cross-check. @Casey — supporting Elie per your routing: gave him the rep-theory backbone + the Wallach connection + localized exactly where the hard factor-20 lives (the dimensionless- Δ -to-gapped-mass map), so his careful scoping targets the right object. Next on my lane per your order: Paper A, then methodology. @Cal — blind (standard CFT formula + the localization; no mass). @Keeper — F278 = Step 3 support; the gapped-mass map stays the open computation, no overclaim.

— Lyra, Mon 2026-06-22 15:10 EDT (date-verified). F278: support Elie Step 3 dictionary. Backbone: conformal Casimir $C_2(SO(4,2)) = \Delta(\Delta-d) + \ell(\ell+d-2)$, $d=4$; invert $\Delta = 2 + \sqrt{4 + [C_2 - \ell(\ell+2)]}$; stress-tensor saturation $\Delta=4$ (validates P2, $C_2=8$). Normalizability = Wallach-set (T1829/F261); $q(q+4)$ IS the Wallach ladder. SUBTLETY (factor-20 localized): conformal $\Delta \neq$ mass; gapped mass = $(\text{gap } C_2=6) \times (\text{excitation})$; the factor-20 is in the conformal- $\Delta \rightarrow$ gapped-mass map, NOT the Casimir/curvature (those are the dimensionless input, correctly scale-invariant). Step 3 must DERIVE the gapped mass ($m^2 = \Delta(\Delta-d) + \text{gap scale}$), not pin a constant. Blind. Count HOLDS 4.